

GEE variables for SDM

Gengping Zhu

gengping.zhu@wsu.edu

Correlative approach:



openModeller

<http://openmodeller.sourceforge.net>

GEE intro

- <https://earthengine.google.com/>
- Here, I am providing two approaches for using GEE to process data for our SDM

Google Earth Engine is a cloud-based platform for analyzing and visualizing planetary-scale geospatial and satellite data.  Google for Developers +2

What It Does

- **Massive Data Catalog:** Stores over 90 petabytes of analysis-ready satellite imagery and geospatial datasets (like Landsat, Sentinel, and weather data) going back more than 50 years.  Google Cloud
- **Cloud Computing:** Uses Google's massive infrastructure to process and analyze large geographic datasets quickly without needing local high-performance hardware.  Geohackweek +1
- **Real-World Tracking:** Helps scientists and developers detect changes, map trends, and study deforestation, drought, and climate monitoring.  Google Earth Engine +1

How Users Access It

- **Development:** Provides Python and JavaScript APIs via a web-based Code Editor or Google Colab.
- **Cost:** Remains free for academic, research, and non-profit use, with paid options available for commercial projects.  Google Earth Engine +2

Approach I: GEE in QGIS

Qiusheng Wu

Building open-source tools for geospatial data science and GeoAI.



Associate Professor | Amazon Scholar

[Department of Geography & Sustainability](#),
University of Tennessee, Knoxville

[309 Burchfiel Geography Building](#),
Knoxville, TN 37996-0925

qwu18@utk.edu | [gishub.org](https://github.com/qwu18)

Research Interests: Geospatial Data Science,
GeoAI, Remote Sensing, Open-Source
Software Development, Cloud Computing,
Wetland Hydrology

[CV \(PDF\)](#) | [Google Scholar](#) | [ResearchGate](#)
| [ORCID](#) | [LinkedIn](#) | [YouTube](#) |
[Twitter](#) | [Facebook](#) | [GitHub](#) | [Bluesky](#)
| [OpenGeoPub](#) |

<https://github.com/opengeos/qgis-gee-data-catalogs-plugin>

<https://www.youtube.com/watch?v=nZ3D6wLKJQw>

Earth Engine Data Catalogs Plugin for QGIS

A comprehensive QGIS plugin for browsing, searching, and loading Google Earth Engine data catalogs directly in QGIS.

QGIS Plugin

License MIT

Features

- **Dynamic Data Catalog Browser:** Automatically loads datasets from official GEE catalog and community datasets
 - [Official Earth Engine Data Catalog](#) - 780+ datasets
 - [Awesome GEE Community Catalog](#) - 4,360+ community datasets
- **Search & Filter:** Search datasets by keywords, tags, providers, data types, and sources
- **Ranked Search & Saved Workflows:** Relevance-ranked search, favorites, recent datasets, and reusable analysis recipes
- **Persistent Catalog Cache:** Disk-backed catalog cache for faster startup and offline fallback
- **Advanced Filtering for ImageCollections:**
 - Date range filtering
 - Bounding box (current map extent)
 - Cloud cover percentage
- **Time Series:** Create time series layers from ImageCollections
- **Pixel Time Series Inspector:** Click on the map to extract and chart time series at a point (similar to Earth Engine's `ui.Chart.image.series`)
- **Flexible Image Loading:**
 - Load composite images (mosaic, median, mean, min, max)
 - Select and load individual images from ImageCollections
- **Code Console:** Write and execute geemap/Earth Engine Python code directly

80 Petabytes Satellite Data For Free!

QGIS

QGIS Data Catalogs

Earth Engine Data Catalogs

to Refresh

Browse Search Time Series Load Code Conversion Inspector Export

Source: All Catalogs

Dataset	Type	Source
> Landsat (148)		
> Sentinel (28)		
> MODIS (132)		
> Elevation (525)		
> Land Cover (126)		
> Climate & Weather (339)		
> Boundaries (211)		
> Night Lights (38)		
> Population (50)		
> Water (580)		
> Vegetation (213)		
> Atmosphere (100)		
> Agriculture (22)		
> Soil (120)		
> Fire (56)		
> Ocean (41)		
> Urban (1315)		
> Other (1002)		

Dataset Information

QGIS

Name: ESA WorldCover 10m v200
ID: ESA/WorldCover/v200
Type: ImageCollection
Provider: ESA WorldCover Consortium



Description:
ee.ImageCollection('ESA/WorldCover/v200')...
Start Date: 2021-01-01
End Date: 2022-01-01
Keywords: esa, landcover, landuse, landuse-landcover, sentinel1-derived, sentinel2-derived

Add to Map Configure & Add

Time Series

Time-series Satellite Images

The screenshot displays the QGIS desktop environment. The main map area is split vertically to compare two satellite images of the same region. The left image, representing the year 2013, is rendered with a false-color composite where vegetation is shown in bright green, water in dark blue, and bare soil or urban areas in brownish-yellow. The right image, representing the year 2025, uses a different color scheme, with vegetation in red, water in dark blue, and land in shades of orange and brown. A vertical blue line serves as a swipe tool to toggle between the two images. On the left side, the 'Layers' panel lists the data sources: 'HLSS30: HLS-3 LandSat Operational Lan', '2025', '2013', 'LandSat_013_2025', 'Band 1 (Grayscale)', 'Band 2', 'Band 3', 'LandSat_013_2025', 'Band 1 (Grayscale)', 'Band 2', 'Band 3', and 'OpenStreetMap'. On the right side, the 'Earth Engine Data Catalog' panel is open, showing 'Property Filters (Optional)' with filters for 'SUN_ELEVATION > 30', 'SUN_AZIMUTH < 180', and 'MOIST_TILE == 10000'. Below this, the 'Time Series Options' section shows 'Frequency: year' and 'Reducer: median'. The 'Visualization Parameters' section shows 'Bands: B6, B5, B4', 'Min: 0', 'Max: 0.4', and 'Palette: e.g., blue, green, red, or white'. A 'Time Slider' is set to '2013 (1/13)'. At the bottom right, the 'Layer Swipe Tool' panel is active, showing 'Layer Selection' with 'Left Layer: 2013' and 'Right Layer: 2025', 'Swipe Direction' set to 'Horizontal', and 'Swipe Position' at 50%. The status bar at the bottom right indicates the time as 8:46 AM on 1/21/28.

Earth Engine Data Catalogs

Refresh

Browse

Search

Time Series

Load

Code

Conversion

Inspector

Export

Source:

Community Catalogs

Dataset	Type	Source
▸ Landsat (38)		
▸ Sentinel (13)		
▸ MODIS (21)		
▸ Elevation (418)		
▸ Land Cover (88)		
▸ Climate & Weather (215)		
▸ Boundaries (169)		
▸ Night Lights (18)		
▸ Population (24)		
▸ Water (554)		
▸ Vegetation (154)		
▸ Atmosphere (96)		
▸ Agriculture (20)		
▸ Soil (99)		
▸ Fire (46)		
▸ Ocean (23)		
▸ Urban (1315)		
▸ Other (1050)		



Approach II: Link AI with GEE

- I can not write Java script, but I still need GEE to process my data
- Let's try ChatGPT and Claud

write the GEE Code Editor script to download bio1, bio5, and bio6 in WorldClim BIO Variables V1 from Google Earth Engine (GEE) for Guangdong Province, China, please download as multi-Band Raster at 10 km resolution.

- write the GEE Code Editor script to download Daymet daily tmax and tmin temperature data from Google Earth Engine (GEE) for this area [-121.019709, 44.625742989505234, -117.580413, 48.54458101049476], please download as multi-Band Raster

Demo

Google Earth Engine

Search places and datasets...

Inspector Console Tasks Ask

Use `print(...)` to write to this console.

The Annual Developer Satisfaction Survey is live!
Take 10 - 15 minutes to answer the survey by [clicking here](#). Your feedback will help us to make changes and improve Google Earth Engine.

Scripts Docs Assets

Filter scripts... NEW

Owner (2)

- users/gengpingzhu/Daymet
 - Cropland
 - DynamicWorld
 - MutipleYear
 - tmax_tmin
- users/gengpingzhu/EVI
 - MutipleYear

Writer

Reader (1)

- users/gisizw/GEE-PICX

Archive

No accessible repositories. Click

New Script

Get Link Save Run Reset Apps

1

Map Satellite